

Discrete Mathematics and graph Theory

Chapter Three: Elements of Graph Theory

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2024/25

Out lines

In this chapter we will discuss the following topics:

- 3.1.** Definition and examples of a graph
- 3.2.** Matrix representation of a graph
- 3.3.** Isomorphic graphs
- 3.4.** Path and connectivity of a graph
- 3.5.** Complete, regular and bipartite graphs
- 3.6.** Eulerian and Hamiltonian graphs
- 3.7.** Trees and forests (Rooted and Binary trees)
- 3.8.** Planar graphs
- 3.9.** Graph coloring

Elements of graph Theory

Graph theory is a branch of mathematics that deals with arrangements of certain objects and relation ships between these objects. Graph theory is broadly classified into two: **nondirected** graphs and **directed** graphs (digraphs). The main elements of graph theory in discrete mathematics are vertices (or nodes) and edges (or links), which are used to model relationships between objects.

The following are some examples dealing with real life situations.

1. Electrical network
2. Telephone communication
3. Computer flow chart
4. Organization chart

Definition

A graph G is a discrete structure defined as an ordered pair (V, E) , where $V = V(G)$ is a non-empty set of vertices (or nodes or points) and $E = E(G)$ is a set of edges (or links/lines) connecting pairs of vertices.

Notation: Edges consisting of vertices u and v is denoted by $e = (u, v)$.

- Graphs are represented pictorially. Each vertex v in V is represented by a dot (small circle), and edge $e = (v_1, v_2)$ is represented by a curve which connects its end points v_1 and v_2 .

Example

In the graph G below, find V and E .

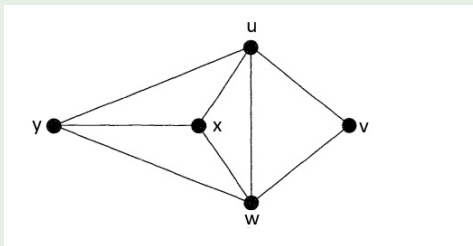


Figure: Pictorial representation of graphs G .

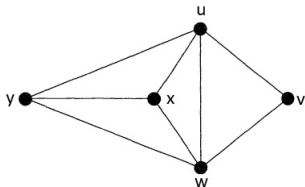
Definition

Two or more edges joining the same pair of vertices are called **multiple (parallel)** edges.

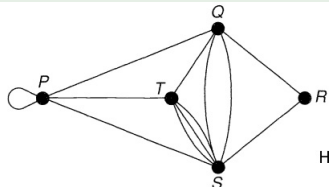
An edge joining a vertex to itself is called a **Loop**.

Example

Determine the loops and multiple(parallel) edges of the following non directed graphs.



(a)



(b)

Figure: (a) Graph G, (b) Graph H

Depending on the presence or absence of loops and multiple edges, graphs are generally divided into the following classes.

Definition

Simple graph: A graph is simple if it has no loops and no multiple edges

Multigraph: A graph which consists of parallel (multiple edges) is called a multigraph.

Example

In the previous example, G is a simple graph, whereas H is a multigraph.

Definition

The order of a graph G is the number (cardinality) of $V(G)$, denoted $|V(G)|$ or $|G|$. The size of a graph G is the number (cardinality) of $E(G)$, denoted $|E(G)|$ or $||G||$.

Example

Determine the order $|G|$ and the size $||G||$ of the graph G .

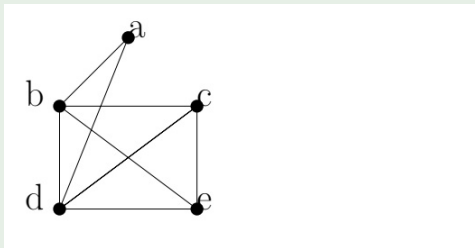


Figure: Graph G .

Definition

- Two vertices u and v are adjacent if they are connected by an edge. The vertices u and v are then incident with such an edge.
- Two edges are said to be adjacent if they have a vertex in common.

Example



adjacent vertices



adjacent edges

Degree of a vertex

Definition

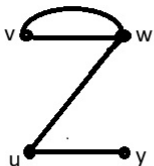
The degree of a vertex v in a non directed graph G , denoted by $\deg(v)$, is

- (i) the number of edges in G which contain V (incident with v) if it has no loops.
- (ii) the number of edges in G which contain V plus twice the number of loop(s) if it has loop(s).

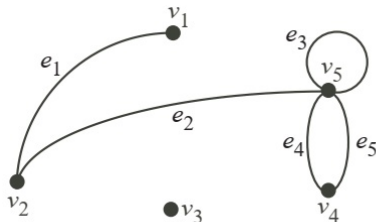
☛ By convention, a loop at a vertex v contributes 2 (rather than 1) to the degree of v .

Example

Determine the degree of each of the vertices of the following non directed graphs.



(a)



(b)

Figure: (a) Graph G, (b) Graph H

Definition

A vertex v in the graph G is said to be

- (a) even if $\deg(v)$ is even, and odd if $\deg(v)$ is odd.
- (b) isolated if $\deg(v) = 0$.
- (c) pendant if $\deg(v) = 1$.

Example

Consider the graph H above.

Minimum and Maximum degree:

The minimum degree of G is $\delta(G) = \min\{\deg(v) : v \in V\}$. The maximum degree of G is $\Delta(G) = \max\{\deg(v) : v \in V\}$.

Example

Find the minimum and maximum degree of G given below.

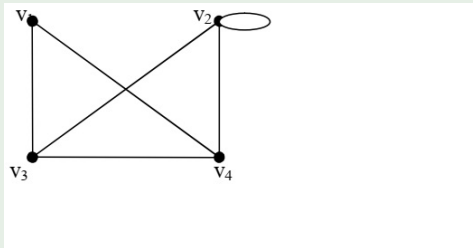


Figure: Graph G .

Remark

The degree sequence of a graph consists of the degrees written in increasing order, with repeats where necessary.

Example

Determine the degree sequence of the graph G below.

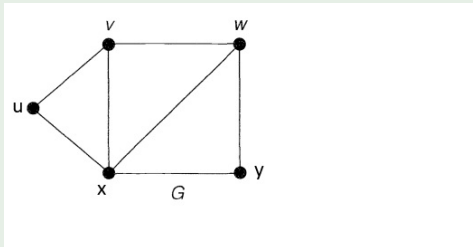


Figure: Graph G .

Theorem (Handshaking Lemma)

Let $G = (V, E)$ be a graph. Then,

$$\sum_{i=1}^n \deg(v_i) = 2|E|$$

Example

Consider the above graph.

Corollary

In any graph G , the number of vertices of odd degree is even.

Example

Is there a simple graph with degree sequence

- ① $(1, 2, 3, 4, 5)$?
- ② $(1, 1, 3, 3, 3, 4, 6, 7)$?

Matrix representation of Graphs

Graphs can be represented by any one of the following types of matrices.

Adjacency matrix: a matrix that describes the adjacency relationships between vertices of a graph.

Incidence Matrix: - A matrix that describes the incidence relationships between vertices and edges of a graph.

Definition

Suppose G is a graph with m vertices and suppose the vertices are ordered as $v_1, v_2, \dots, v_m$. Then the adjacency matrix $A = (a_{ij})$ of the graph G is the $m \times m$ matrix defined by:

$$a_{ij} = \begin{cases} n, & \text{if there are } n \text{ edges joining } v_i \text{ and } v_j \\ 0, & \text{otherwise,} \end{cases}$$

Example

Determine the adjacency matrix for the graph G below.

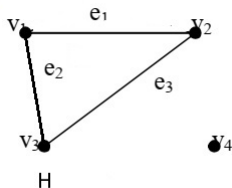
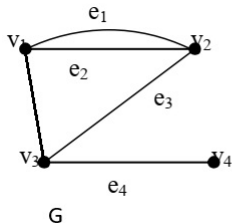


Figure: (a) Graph G , (b) Graph H

Note

- Adjacency matrix is symmetric.
- If the entries in a vertex is zero, then the vertex is isolated.

In the adjacency matrix of any graph G , the degree of each vertex v_i is

- (a) the sum of all entries in the i^{th} row (or i^{th} column if there is no loop at the vertex v_i).
- (b) row(or column) sum plus the number of loops if there is (are) loop(s) at v_i .

Example

Given that the matrix below is the adjacency matrix of the graph G .

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- (a) find the degree of each vertex.
- (b) Determine the number of edges of the graph G .
- (c) Draw the graph G .

In a simple graph:

- ✈ All entries of the adjacency matrix are 0 or 1 (since there either is or is not an edge between each pair of vertices).
- ✈ The diagonal is all 0's (since there are no loops).

Incidence Matrix

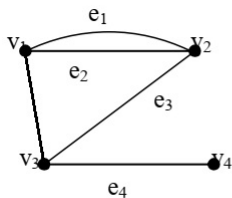
Definition

Suppose G is a graph with vertices $v_1, v_2, \dots, v_m$. The incidence matrix $I = (b_{ij})$ of the graph G is the $m \times n$ matrix defined by:

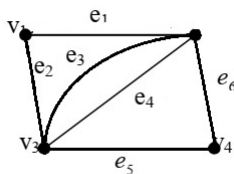
$$b_{ij} = \begin{cases} 1, & \text{if } e_j \text{ is incident on } v_i \\ 0, & \text{otherwise,} \end{cases}$$

Example

Determine the incidence matrices for the graph G and H below.



G



H

Isomorphism of Graphs

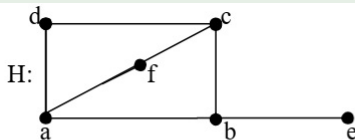
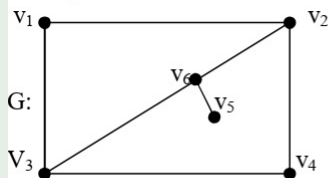
It is possible for two graphs look different but to represent the same.

Definition

Graphs $G(V, E)$ and $G^*(V^*, E^*)$ are said to be isomorphic if there exists a one to one correspondence $f : V \rightarrow V^*$ such that (u, v) is an edge of G iff $(f(u), f(v))$ is an edge of G^* .

Example

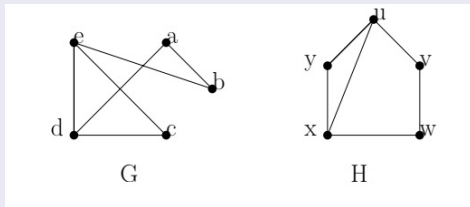
Show that G and H are isomorphic graphs.



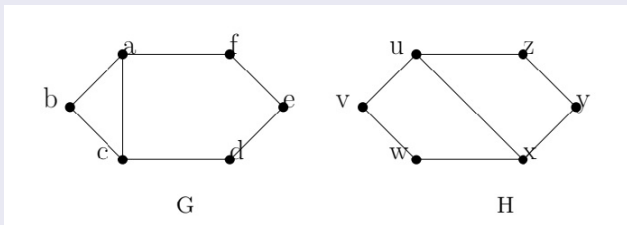
- 1 Two graphs $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ are said to be isomorphic if a 1 – 1 correspondence f exists from V_1 to V_2 such that u and v are adjacent in G_1 iff $f(u)$ and $f(v)$ are adjacent in G_2 .
 - 2 If $G \cong H$, then the degrees of corresponding vertices are equal i.e., degrees are preserved.
 - 3 Two isomorphic graphs must have the same number of vertices and edges.
- ☛ We can check by using adjacency matrix, if the adjacency matrix of the two graphs are the same, then we can say the two graphs are isomorphic.

Exercise

- 1 Show that the following two graphs are isomorphic.



- 2 Show that the following two graphs are not isomorphic.



Subgraph

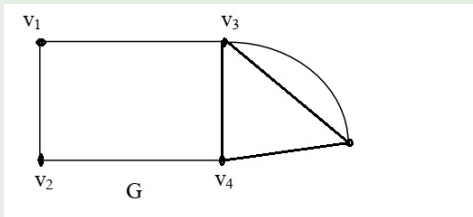
Definition

A graph H is called a subgraph of G if every vertex of H is also a vertex of G and every edge of H is also an edge of G .

Symbolically H is a subgraph of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

Example

Draw the subgraphs of the graph G below.



Note

- ① A graph with $V \neq \{\}$ but with no edge ($E = \{\}$) is null graph.
- ② A graph and its null graph are trivial subgraphs.
- ③ A subgraph H of G is called a spanning subgraph of G iff $V(H) = V(G)$.
- ④ A subgraph H of G is called a proper subgraph if $H \neq G$.

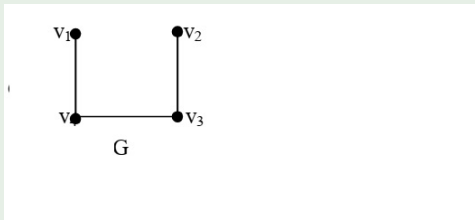
Complement of a graph

The complement of G is a simple graph $\bar{G}$ such that:

- (i) Vertices of G and $\bar{G}$ are the same,
- (ii) $(u, v) \in E(\bar{G}) \Leftrightarrow (u, v) \notin E(G)$.

Example

Draw the complement $\bar{G}$ of the graph G below.



Paths and connectivity

Definition

A walk in G is a nonempty sequence $W = v_0 e_1 v_1 e_2 v_2 \cdots e_n v_n$ whose terms are alternately vertices and edges, such that $e_i = v_{i-1} v_i$ for every $i, 1 \leq i \leq n$.

We say W is a walk from v_0 to v_n (or a (v_0, v_n) -walk). The vertices v_0 and v_n are called the ends of W .

✈ The number of edges in a walk is called the length of the walk.

Definition

Let $W = v_0 e_1 v_1 e_2 v_2 \cdots e_n v_n$ be a walk. If all vertices in W are distinct, we call W a path.

- ✉ When the graph is simple, we denote this path by its vertex sequence $P = v_0 v_1 \cdots v_n$ (because listing these vertices uniquely determines the path).

Given a path $P = v_0 e_1 v_1 e_2 v_2 \cdots e_{n-1} v_{n-1} e_n v_n$

- ☞ P is said to traverse edges $e_1, e_2, \cdots, e_n$, and visit $v_0, v_1, v_2, \cdots, v_n$.
- ☞ v_0 is the initial vertex and v_n is the terminal vertex.
- ☞ A path is said to be closed if $v_0 = v_n$.
- ☞ P is said to be open if $v_0 \neq v_n$.
- ☞ P is called a cycle if it is a closed simple path (i.e., all the vertices are distinct except v_0 and v_n).
- ☞ P is called a trail if all the edges are distinct (i.e., a path that does not traverse the same edge more than once)
- ☞ A simple path of length ≥ 1 with no repeated edges and whose end points are equal is called a circuit.

Note

A loop is a cycle of length 1.

Example

Consider the graph G below. Give your own walks, paths, cycles and trials of the graph G .

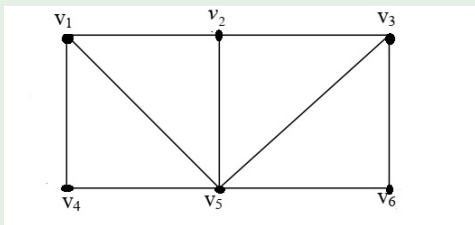


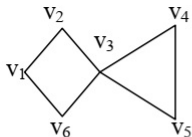
Figure: Graph G .

Definition

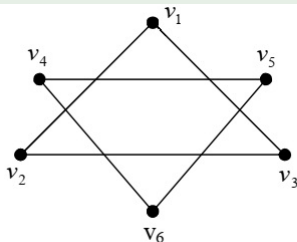
A graph G is connected if there is a path between any two of its vertices. A graph is called disconnected if it is not connected.

Example

Are the graphs G and H given below connected?



(a) Graph G



(b) Graph H

Theorem

A connected graph with n -vertices must have at least $n - 1$ edges. That is, $|E| \geq 1$.

Example

Theorem

- (a) *A connected graph with 2 vertices has at least $(2 - 1) = 1$ edge.*
- (b) *A connected graph with 3 vertices has at least $(3 - 1) = 2$ edge.*

Complete, regular and Bipartite Graphs

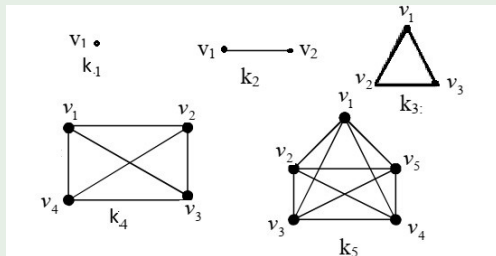
Definition

A graph G is said to be complete if every vertex in G is connected to every other vertex in G .

Notation: The complete graph with n -vertices is denoted by k_n .

Example

The following are some of the complete graphs.



Question: How many edges are there in K_n ?

Note

In a complete graph k_n , there are $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.

 All vertices of a complete graph are mutually adjacent.

Example

Draw a complete graph k_6 .

Question: How many edges are there in K_n ?

Note

In a complete graph k_n , there are $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.

☞ All vertices of a complete graph are mutually adjacent.

Example

Draw a complete graph k_6 .

Regular graphs

Definition

A graph G is said to be regular of degree k or k -regular if every vertex has degree k .

In other words, a graph is regular if every vertex has the same degree.

Example

The following are some examples of regular graphs.

 0-regular

 1-regular



or



or



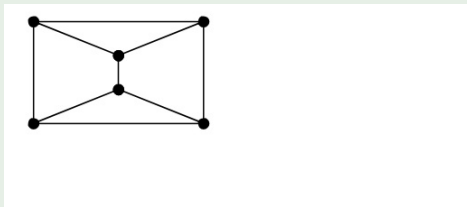
2-regular

Theorem

A complete graph K_n is a regular graph of degree $n - 1$.

Example

The following graph is 3-regular but not complete.



Bipartite graph

Definition

A graph G is said to be bipartite if its vertices V can be partitioned into two subsets M and N such that each edge of G connect a vertex of M to a vertex of N .

That is, none of the edges in G connect vertices with in the same set M or N .

Complete Bipartite graph: is a graph for which each vertex of M is connected to each vertex of N ,

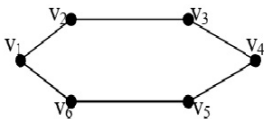
Notation: A complete bipartite graph is denoted by $K_{m,n}$ where m is the number of vertices in M and n is the number of vertices in N ($m \leq n$).

Example

Draw the Complete bipartite graphs $K_{1,5}$, $K_{2,4}$, $K_{3,3}$.

Exercise

- 1 What will be the number of edges of a complete bipartite graph $K_{m,n}$?
- 2 Is the graph given below bipartite? or complete bipartite?



Eulerian and Hamiltonian Graphs

Definition

Eulerian Path:

A path that traverse every edges of a graph exactly once is called an Eulerian path (Eulerian).

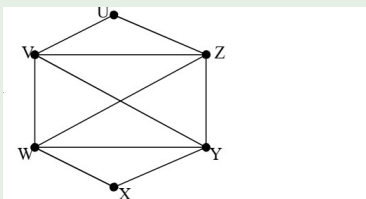
Eulerian Cycle:

An Eulerian path that begins and ends at the same vertex is called an Eulerian trial (cycle).

A graph that contains an Eulerian cycle is called an Eulerian Graph.

Example

Consider the graph G given below.



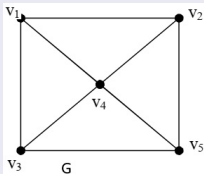
- (a) Show that G has an Euler closed trial.
- (b) Find an Euler trial starting and ending at U .

Theorem (Necessary and sufficient condition)

A connected graph is Eulerian iff all of its vertices have even degree.

Exercise

Explain why the graph G given below is not Eulerian.



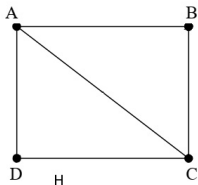
Theorem

A connected graph G with 3 or more vertices contains an Eulerian path but not an Eulerian cycle if and only if G has exactly two vertices with odd degrees.

- ☞ The Eulerian path in the graph G begins at one odd vertex and ends at another odd vertex.

Example

Determine the Eulerian path (if any) for the graph H below.

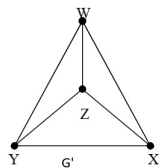
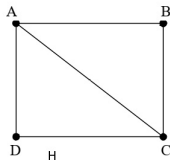
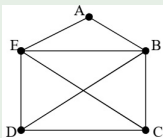


Remark

- (i) A connected graph G with exactly two odd vertices is called *Semi-Eulerian graph*.
- (ii) A graph (connected) with more than two odd vertices is neither Eulerian nor semi-Eulerian.

Example

Determine whether the following graphs are Eulerian, semi-Eulerian or neither.



Hamiltonian Graphs

Definition

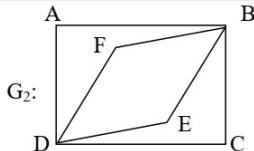
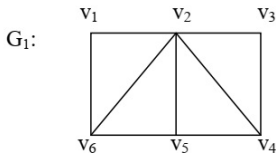
- ✈ a path that visits every vertex in a graph G exactly once is called a Hamilton path.
- ✈ A closed Hamilton path is called a Hamilton cycle.
- ✈ G is called a Hamiltonian graph if it admits a Hamiltonian cycle.

Remark

An Eulerian trail traverses every edge exactly once, but may repeat vertices while a Hamiltonian cycle (circuit) visits each vertex exactly once but may repeat edges.

Example

- 1 Draw a Hamilton graph having Hamilton cycle that traverse an edge more than once.
- 2 Consider the graphs G_1 and G_2 below.



Which of the following is true about the graphs?

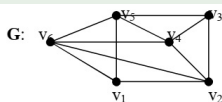
- (a) G_1 is Hamiltonian but not Eulerian.
- (b) G_2 is Eulerian but not Hamiltonian.
- (c) G_1 is both Eulerian and Hamiltonian.
- (d) G_1 and G_2 are Eulerian and Hamiltonian.

Theorem

If G is a simple graph with vertices $n \geq 3$ and if $\deg(u) + \deg(v) \geq n$ for all pairs of non-adjacent vertices u and v , then G is Hamiltonian. (The converse is not always true)

Example

Show that a graph G given below is Hamilton.

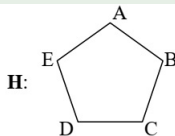
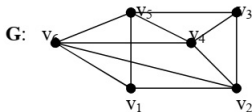


Corollary

If G is a simple graph with number of vertices $n \geq 3$ and if $\deg(v) \geq \frac{n}{2}$ all vertices v in G , then G is Hamiltonian.

Example

The graph G below is Hamiltonian (Verify based on the corollary given above). Give one Hamiltonian cycle. What about the graph H ?



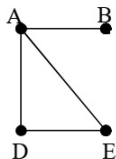
Question: What do you observe the from the converse of the corollary?

Trees and Forests

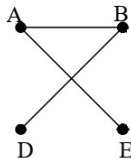
☞ A graph without cycle is said to be cycle free (acyclic) graph.

Example

Observe the following graphs.



Graph G



Graph H

Graph G is Cyclic where as graph H is acyclic.

Tree

Definition

A tree is connected graph with no simple circuit (cycle).

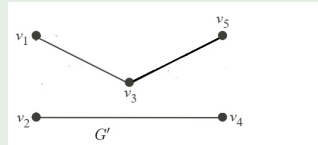
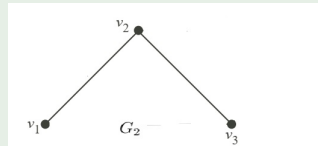
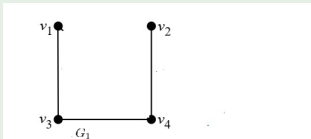
Forest

Definition

A graph with no cycle (not necessary connected) is called a forest.

Example

Classify the following graphs as Tree or Forest.



Remark

- ☞ The tree consisting of single vertex with no edge is called a degenerate tree.
- ☞ The vertices of degree 1 in a tree are called leaves, and the edges of a tree are called branches.

Exercise

Sketch all the trees T with exactly 6 vertices. What is the number of edges in these trees?

Theorem

Let G be a graph with $n > 1$ vertices. Then the following are equivalent.

- (i) G is a tree
- (ii) G is cycle-free and has $n - 1$ edges
- [iii] G is connected and has $n - 1$ edges

Example

A tree has 4 vertices of degree 3 and 3 vertices of degree 2. The remaining vertices have degree 1. Find the total no of vertices in the tree.

Answer: $n = 13$.

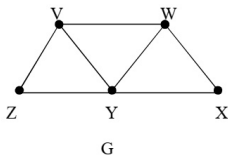
Spanning trees

Definition

A spanning tree of a connected graph G is acyclic connected subgraph of G which contains all the vertices of G .

Example

Draw the spanning trees of the graph G below.



Planar graphs

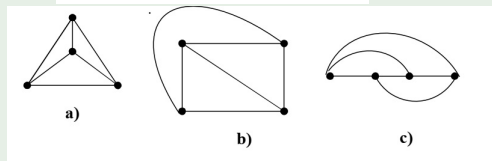
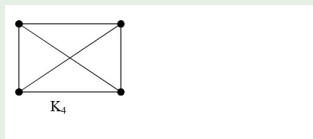
Definition

A graph G which can be drawn in a plane with its edges intersecting only at (common) vertices only is called a planar graph.

- ☞ A graph that has no such plane representation (or depiction) is called a non-planar graph.

Example

- ① K_4 is a planar graph. The graph K_4 given in figure below has planar depictions (or plane drawings) shown in figures a , b and c.



- ② A complete graph k_5 and a complete bipartite graph $k_{3,3}$ are non planar. Draw the plane representations of the graphs. What can we say about a complete bipartite graph $k_{2,3}$?

Definition

If G is a planar graph, then the plane representation of G splits the plane into regions called faces of G .

The unbounded region outside the graph is called the infinite face.

Degree of the faces

Definition

If f is any face of a graph, then the degree of f , denoted by $\deg(f)$, is the number of edges encountered in a walk (or path) that begins and ends at the same vertex around the boundaries of the face f . If all faces have the same degree r , then G is a face-regular graph of degree r .

Example

Determine the degree of faces of the plane representation of k_4 .

Theorem

The sum of all the degrees of the faces in a planar graph is twice the number of edges in the graph. That is,

$$\sum_{i=1}^n \deg(f_i) = 2|E|$$

Euler's Formula

Theorem

For any connected planar graph G , $|V| - |E| + |F| = 2$ Where $|V|$ denotes the number of vertices, $|E|$ the number of edges and $|F|$ the number of faces in G .

Example

Verify Euler's formula for the planar graph K_4 .

Graph Coloring

Definition

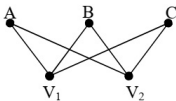
Consider a graph G . a vertex coloring or simply a coloring of G is an assignment of colors to the vertices of G such that adjacent vertices have different colors.

For a positive integer K , a graph G is said to be K -colorable or K -colored if there exists a coloring of G which uses K colors.

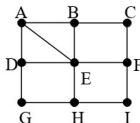
Example

In each of the graphs below, determine the number of colors needed to color the vertices of the graph in such a way that no pair of adjacent vertices have the same color.

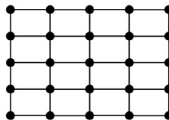
a)



b)



c)

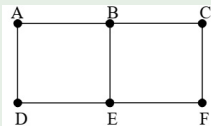


Definition

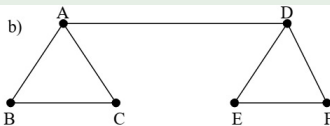
The chromatic number of graph G , denoted by $\chi(G)$, is the minimum number of colors needed to paint (or color) G so that no two adjacent vertices receive the same color.

Example

Find the chromatic numbers of the graphs G and H below.



G



H